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AstroLogic

*A Prolog-Driven Symbolic Reasoning System for Zodiac Profiling,
Tarot Interpretation, and Compatibility Analysis*

Submitted by Group – 6

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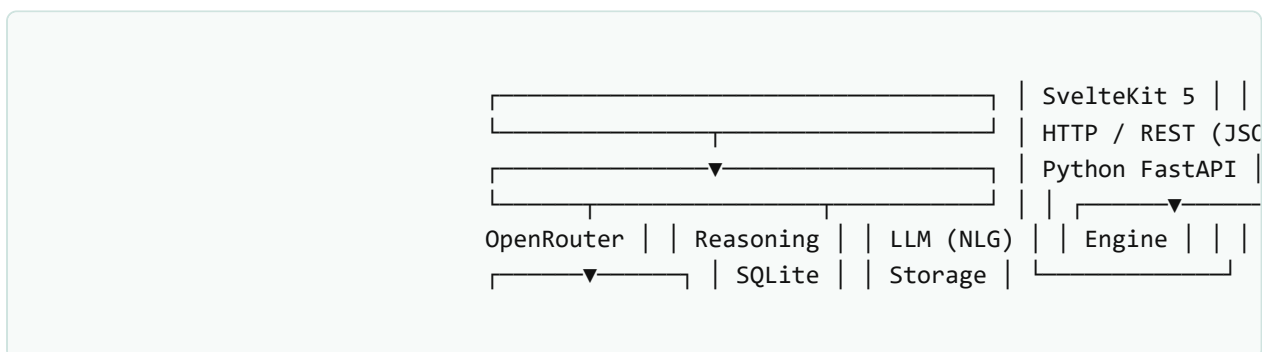
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1. Introduction

AstroLogic is a full-stack, AI-assisted web application that reinterprets Western zodiac astrology and tarot practice as a **symbolic knowledge-representation and reasoning problem**. Rather than treating horoscopes and tarot readings as arbitrary text generated by a language model, AstroLogic encodes the domain – zodiac signs, elements, modalities, ruling planets, personality traits, the 78-card tarot deck, spreads, and compatibility rules – as a formal **Prolog knowledge base**. A large language model (LLM) is used only at the very last step, to translate the structured symbolic output produced by Prolog into natural, readable language. This design keeps the "reasoning" auditable, deterministic, and explainable, while still producing an engaging, conversational user experience.

The system is organized as a three-tier architecture:

- **SvelteKit 5 frontend** (TypeScript, Tailwind CSS) – the interactive web client, including a dashboard, tarot reading flow, horoscope generator, compatibility checker, an AI chat assistant, and a reasoning-trace visualizer.
- **Python FastAPI backend** – the orchestration and API layer that classifies user questions, queries the Prolog engine, calls the OpenRouter LLM service, and persists history to SQLite.
- **SWI-Prolog knowledge base** – the symbolic reasoning core: zodiac facts, tarot facts, and four dedicated reasoning modules that derive profiles, select cards, analyze readings, and score compatibility.



The core design principle is a strict **separation of concerns**: Prolog decides *what* the system reasons (facts, rules, and inference), Python decides *how* the system orchestrates that reasoning, the LLM decides *how* the result is expressed in natural language, and SvelteKit decides *how* the user experiences all of it. If the LLM provider is unreachable, the application degrades gracefully to deterministic, Prolog-derived interpretations rather than failing outright.

AstroLogic is framed explicitly as a **reflective / entertainment** tool. The system prompt and UI copy avoid claims of scientific prediction or medical/psychological diagnosis; the Prolog rules model traditional astrological/tarot symbolism, not empirical claims about personality or the future.

2. Objectives

The objectives below define the scope of Version 1 (V1) of AstroLogic.

1. **Model zodiac knowledge symbolically.** Represent all 12 zodiac signs with their elements, modalities, ruling planets, symbolic traits, and date ranges as Prolog facts, and derive strengths, challenges, and full personality profiles through inference rather than hard-coded text.
2. **Model the tarot deck symbolically.** Represent the full 78-card deck (22 Major Arcana + 56 Minor Arcana) with keywords, upright/reversed meanings, and thematic categories, and use these facts to drive card selection.
3. **Provide context-aware card selection.** Classify a user's free-text question into a category (e.g. love, career, decision) and use Prolog rules to filter eligible cards and recommend an appropriate spread (single card, three-card, Celtic Cross, decision spread, etc.).
4. **Produce explainable readings.** Analyze a set of drawn cards for recurring themes, dominant elements/suits, and conflicting or complementary themes, and expose the underlying rule firings as a **reasoning trace** the user can inspect.
5. **Score zodiac compatibility transparently.** Compute a weighted compatibility score between two signs from four independent factors (element, modality, trait, and planetary symbolism) rather than a single opaque lookup table.
6. **Layer natural-language generation on top of symbolic output.** Use an LLM (via OpenRouter) purely as a presentation layer that narrates Prolog's structured facts, with a deterministic fallback when the LLM is unavailable.
7. **Deliver the reasoning through a usable web application.** Build a SvelteKit interface (dashboard, zodiac explorer, tarot reading, horoscope, compatibility, chat, history, analytics) backed by a documented REST API.

3. Requirements

3.1 Functional Requirements

ID	Requirement	Description
FR-01	Zodiac profile lookup	User selects/enters a birth date; system returns sign, element, modality, ruling planet, traits, strengths and challenges.
FR-02	Tarot card draw	System draws N cards for a chosen spread and returns card identity, orientation, and position-aware meaning.
FR-02a	Custom-count draw	User picks any card count from 1–10 for an open, non-fixed-layout spread instead of a predefined one.
FR-02b	Manual card selection	User hand-picks specific cards (and orientation) directly from the 78-card deck; the reading is generated from exactly that selection instead of a random draw.
FR-03	Question classification	Free-text question is classified into a category to drive spread recommendation and card eligibility.
FR-04	Reading analysis	System detects dominant themes, elements, arcana balance, and theme conflicts across a multi-card reading.
FR-05	Horoscope generation	System generates a themed daily horoscope (theme, guidance, reflection, opportunity, caution) from zodiac + mood input.
FR-06	Compatibility analysis	System scores compatibility between two zodiac signs with a full component breakdown and explanation.
FR-07	AI chat assistant	Conversational endpoint answers questions about signs, cards, and readings using Prolog context + LLM narration.

ID	Requirement	Description
FR-08	Reasoning-trace visualization	Every Prolog inference exposes a structured trace (rule, input, result, explanation) renderable in the UI.
FR-09	History & persistence	Readings and chat sessions are stored in SQLite and retrievable/deletable via the History page.
FR-10	Graceful LLM fallback	If OpenRouter is unreachable or times out (30s), the system serves deterministic Prolog-derived text.
FR-11	Precise natal chart calculation	Given birth date, time, and (optionally) place, the system computes exact Sun, Moon, Ascendant, Midheaven, and outer-planet positions via the Swiss Ephemeris (NASA JPL DE421 data), including sign degree and timezone resolution; falls back to a date-only sign approximation when place/time is omitted.
FR-12	AI-narrated birth chart explanation	System generates an in-depth, bilingual (English/Myanmar) psychological and astrological explanation of a calculated chart – Big Three synthesis, planetary archetypes, major aspects, and elemental constitution.
FR-13	Tarot card image scan	User photographs or uploads a physical tarot card; a vision-capable LLM identifies the card (with a local filename/keyword fallback when the vision API is unavailable) and the system returns its meaning plus an AI-generated explanation.
FR-14	Bilingual interface (i18n)	The entire UI, including zodiac, tarot, and horoscope content, is available in English and Myanmar (Burmese); the chosen locale persists across sessions via local storage.
FR-15	User profile persistence	User-entered profile details (name, birth date/time/place, preferred zodiac sign) are saved locally and reused to pre-fill readings, horoscope, and birth-chart forms.
FR-16	Deep synastry breakdown	Beyond the headline compatibility score, a dedicated endpoint returns per-component scoring, communication/balance themes, strengths,

ID	Requirement	Description
		challenges, and complementary trait pairs for two signs.
FR-17	Topic classification	Free-text questions can be classified into a tarot topic independently of a full reading, returning the topic and its associated Prolog-derived profile.

3.2 Non-Functional Requirements

- **Explainability** – every symbolic conclusion must be traceable to the specific Prolog rule(s) that produced it.
- **Separation of concerns** – the LLM must never perform core reasoning; it only narrates facts supplied by Prolog.
- **Resilience** – the system must degrade gracefully (fallback templates) rather than fail when the LLM provider is down.
- **Responsiveness** – REST endpoints should respond within a few seconds for typical queries; LLM calls are bounded by a 30s timeout.
- **Maintainability** – the Prolog knowledge base is split into cohesive modules (base facts vs. reasoning modules) to keep rules easy to extend.
- **Ethical framing** – content must present itself as reflective/entertainment, avoiding predictive or diagnostic claims.

3.3 Technology Stack

Layer	Technology
Frontend	SvelteKit 5, TypeScript, Tailwind CSS, Vite, Vitest
Backend / API	Python, FastAPI, Pydantic schemas
Symbolic Reasoning	SWI-Prolog knowledge base (13 modules, ~4,480 lines of rules/facts)
Natural-Language Generation	OpenRouter LLM API (text narration + a vision-capable model for tarot card image identification)
Astronomical Calculation	pyswisseph (Swiss Ephemeris) over the NASA JPL DE421 dataset, timezonefinder + tzdata for birth-place timezone resolution

Layer	Technology
Internationalization	Custom Svelte i18n store with English and Myanmar (Burmese) translation dictionaries
Persistence	SQLite (schema.sql), Docker Compose for local orchestration
Tooling	ESLint, Prettier, pytest, svelte-check

4. Algorithms

4.1 End-to-End Reasoning Pipeline

Every user-facing request (tarot reading, horoscope, compatibility check, chat message) follows the same seven-step pipeline:

1. User input is captured by the SvelteKit frontend and sent to FastAPI over REST.
2. FastAPI performs lightweight NLP to classify the question (e.g. love / career / decision / general).
3. FastAPI's `PrologService` queries SWI-Prolog with the classified inputs.
4. Prolog executes rule-based inference and returns **structured symbolic results** (facts, scores, traces) – never free text.
5. FastAPI forwards the structured results to OpenRouter with a constrained system prompt.
6. OpenRouter returns natural-language narration grounded in those facts (or, on failure, FastAPI substitutes a deterministic template built directly from the Prolog output).
7. The combined response (symbolic data + narrative + reasoning trace) is returned to SvelteKit and rendered to the user.

4.2 Zodiac Profile Derivation

Given a sign, `generate_profile/2` composes four independent derivations – base facts (element, modality, ruling planet, traits), an element-personality mapping, a modality-approach mapping, and a planetary-influence mapping – into one profile term. Strengths and challenges are derived per-trait via `trait_strength/2` and `trait_challenge/2`, so a sign's five traits expand into five strengths and five challenges automatically, without any sign-specific hard-coding. Birth-date → sign mapping (`zodiac_from_date/3`) uses ordered date-boundary rules with cut-fail so exactly one sign is returned per date.

4.3 Question Classification and Card Selection

A free-text question is classified into a category (love, career, decision, general, ...). The category, together with the user's zodiac sign, forms a *context* that is passed through a filtering pipeline: `card_matches_theme/2`, `card_matches_question/2`, `card_matches_element/2`, and `card_matches_sign/2` each contribute a boolean signal; `card_eligible/2` combines them, `select_eligible_cards/3` produces the eligible set, and `card_priority/3` + `rank_cards/3` order candidates before the top N are drawn for the recommended spread.

4.4 Reading Analysis (Multi-Card Interpretation)

`analyze_reading/3` aggregates per-card analyses across the whole spread: it collects each card's themes via `reading_themes/2`, determines the `dominant_theme/2` by frequency, checks every theme pair against a `theme_conflict/3` table (e.g. *action* vs. *patience*, *independence* vs. *cooperation*) to surface tension in the reading, and separately flags complementary themes. Each card position (past, present, future, challenge, guidance, ...) applies its own interpretive lens through `position_meaning_modifier/2`, so the same card yields a different interpretation depending on where it lands in the spread.

4.5 Weighted Compatibility Scoring

`zodiac_compatibility/3` is a genuine multi-factor weighted algorithm rather than a static lookup table:

Factor	Weight	Basis
Element compatibility	35%	Fire/Air vs. Water/Earth affinity table
Modality compatibility	20%	Cardinal / Fixed / Mutable interaction
Trait compatibility	30%	Count of complementary trait pairs between the two signs
Planetary symbolism	15%	Ruling-planet relationship

The four component scores are combined as $\text{Overall} = 0.35 \cdot E + 0.20 \cdot M + 0.30 \cdot T + 0.15 \cdot P$, then thresholded into `high` (≥ 65), `moderate` (≥ 45), or `low` levels. Because every component is independently queryable, the system can also produce a full **score breakdown** and a natural-language **explanation** for transparency.

An earlier version of this scoring gave every component an artificially high floor (e.g. a trait score of at least 40/100 even for two signs sharing *no* traits and *no* complementary traits, and a flat 55/100 default for any unlisted planetary pairing), which mathematically compressed every possible sign pair into a narrow 44–90 band – in practice this meant almost every pairing was reported as “moderate”, with no pair ever reaching “high”. The component-score tables and the classification thresholds above were subsequently rebalanced so that a genuinely poor elemental/modality/trait match can score low and a genuinely strong one can score high, verified by exhaustively scoring all 66 distinct sign pairs and confirming the result spans the full range (roughly 35–90) with a realistic high/moderate/low split.

4.6 Open & Manual Tarot Draws

Beyond the six fixed-layout spreads (one card, three card, decision, self reflection, relationship, career), the system supports two additional draw modes that reuse the same Prolog analysis pipeline (`analyze_reading_oriented/4`, theme/conflict/direction detection) rather than duplicating it:

- **Custom-count draw.** `custom_spread_recommendation/4` generates `N` generic, order-based positions (“Card 1” ... “Card N”) for any seeker-chosen count from 1–10, still grounding the recommendation in the seeker's zodiac element/modality the way a fixed spread does.
- **Manual card selection.** The seeker picks specific cards (and their orientation) directly from the 78-card deck instead of a random draw; the backend builds the same per-card data structure a random draw would have produced, in the exact order and orientation chosen, and feeds it through the identical reading pipeline – themes, dominant theme, direction, advice, conflicts, and the full reasoning trace are all generated from the seeker's own selection.

Supporting either mode surfaced a latent bug in `position_meaning_modifier/2`: it had no catch-all clause, so a position label outside the five fixed-spread names (such as “Card 6” in an open ten-card draw) made `interpret_card_position/4` fail silently for that card – it still appeared in the UI, but received no per-card interpretation. A catch-all clause was added so every drawn or selected card, in any spread, is always individually interpreted.

4.7 Fallback Algorithm

Before any OpenRouter call, FastAPI starts a 30-second timeout. On timeout, network error, or malformed response, the orchestration layer substitutes a template-based generator that consumes the same structured Prolog output the LLM would have received (themes, scores, traits) and fills a deterministic sentence template – guaranteeing the user always receives a complete, non-empty response.

5. Prolog Codes

The knowledge base contains 13 files (~4,480 lines total) organized as two base fact libraries and four functional reasoning modules, orchestrated by a single entry point.

File	Role	Purpose
<code>zodiac.pl</code>	Base facts	12 signs, elements, modalities, ruling planets, traits, date ranges, symbols (110 lines)
<code>tarot.pl</code>	Base facts	78-card deck: arcana, keywords, upright/reversed meanings, themes (470 lines)
<code>zodiac_profile.pl</code>	Module 1	Profile generation, strengths/challenges, birth-date mapping, element/modality compatibility (709 lines)
<code>card_selection.pl</code>	Module 2	Card theming, eligibility filtering, spread selection and ranking (817 lines)
<code>reading_analysis.pl</code>	Module 3	Per-card and multi-card analysis, theme/conflict detection, advice (743 lines)
<code>synastry.pl</code>	Module 4	Weighted compatibility scoring and synastry explanation (495 lines)
<code>horoscope_rules.pl</code>	Support	Mood → theme mapping for horoscope generation (164 lines)
<code>spread_rules.pl</code>	Support	Question classification → spread recommendation (179 lines)
<code>recommendation_rules.pl</code>	Support	Card recommendation heuristics (195 lines)
<code>reasoning.pl</code>	Support	Cross-module orchestration (147 lines)
<code>compatibility.pl</code>	Support	Legacy/simple compatibility helpers (152 lines)
<code>history_analysis.pl</code>	Support	Aggregate analysis over stored reading history (215 lines)

File	Role	Purpose
main.pl	Entry point	Loads all modules; exposes convenience queries (84 lines)

5.1 Base Facts – Zodiac Knowledge

```
% --- Elements ---
element(aries, fire).
element(taurus, earth).
element(gemini, air).
element(cancer, water).
...

% --- Modalities ---
modality(aries, cardinal).
modality(taurus, fixed).
modality(gemini, mutable).
...

% --- Symbolic Traits ---
zodiac_traits(aries, [courage, initiative, independence,
                     impulsiveness, leadership]).
zodiac_traits(taurus, [patience, reliability, sensuality,
                     stubbornness, determination]).
...

% --- Helper: Get all signs of a given element ---
signs_of_element(Element, Signs) :-
    findall(Sign, element(Sign, Element), Signs).
```

Excerpt from `prolog/zodiac.pl` – declarative facts for all 12 signs.

5.2 Module 1 – Zodiac Profile Generation

```
:- use_module(zodiac).

% --- get_element/2 ---
get_element(Sign, Element) :-
    zodiac(Sign),
    element(Sign, Element).

% --- Trait strength / challenge mapping ---
trait_strength(initiative, 'Takes action and leads').
trait_challenge(initiative, 'May act impulsively without thinking').
...

% --- generate_profile/2 ---
% Combines base facts + element personality + modality
% approach + planetary influence into one profile term.
generate_profile(Sign, Profile) :-
    zodiac_profile(Sign, Base),
    element_personality(Sign, ElementDesc),
    modality_approach(Sign, ModalityDesc),
    planetary_influence(Sign, PlanetDesc),
    Profile = Base.put(_{
        element_personality: ElementDesc,
        modality_approach: ModalityDesc,
        planetary_influence: PlanetDesc
    }).
```

Excerpt from `prolog/zodiac_profile.pl` – every trait maps to a strength and a challenge, so profiles are derived, not hand-written per sign.

5.3 Module 2 – Card Theme Classification & Selection

```
:- use_module(zodiac).
:- use_module(tarot).
:- use_module(zodiac_profile).

% --- Card Theme Classification ---
card_theme(the_fool, new_beginnings).
card_theme(the_fool, spontaneity).
card_theme(the_hermit, reflection).
card_theme(the_hermit, solitude).
card_theme(the_hermit, introspection).
...

% --- Eligibility pipeline ---
card_matches_theme(Card, Theme)      :- card_theme(Card, Theme).
card_matches_question(Card, Cat)     :- ... .
card_eligible(Card, Context) :-
    card_matches_question(Card, Context.category),
    ( card_matches_sign(Card, Context.sign) ; true ).

select_eligible_cards(Context, AllEligible, Filtered) :-
    findall(C, card_eligible(C, Context), AllEligible),
    rank_cards(AllEligible, Context, Filtered).
```

Excerpt from `prolog/card_selection.pl` – each of the 78 cards is tagged with multiple themes, then filtered/ranked per question context.

5.4 Module 4 – Weighted Synastry Scoring

```
:- use_module(zodiac).
:- use_module(zodiac_profile).
:- use_module(card_selection).

% zodiac_compatibility/3
% Derived from element + modality + traits + planet,
% NOT a single hardcoded table.
zodiac_compatibility(Sign1, Sign2, Result) :-
    Sign1 \= Sign2,
    element(Sign1, E1), element(Sign2, E2),
    modality(Sign1, M1), modality(Sign2, M2),
    ruling_planet(Sign1, P1), ruling_planet(Sign2, P2),
    element_compatibility_score(E1, E2, EScore),
    modality_compatibility_score(M1, M2, MScore),
    trait_compatibility_score(Sign1, Sign2, TScore),
    planetary_compatibility_score(P1, P2, PScore),
    OverallScore is (EScore * 0.35) + (MScore * 0.20)
                    + (TScore * 0.30) + (PScore * 0.15),
    ( OverallScore >= 65 -> Level = high
    ; OverallScore >= 45 -> Level = moderate
    ; Level = low
    ),
    Result = compatibility_result{
        sign1: Sign1, sign2: Sign2, level: Level,
        score: OverallScore, element_score: EScore,
        modality_score: MScore, trait_score: TScore,
        planetary_score: PScore
    }.

element_compatibility_score(fire, fire, 90) :- !.
element_compatibility_score(fire, air, 85)  :- !.
element_compatibility_score(air, fire, 85)  :- !.
element_compatibility_score(air, air, 80)   :- !.
element_compatibility_score(earth, earth, 85) :- !.
element_compatibility_score(fire, water, 22) :- !. % most friction-prone pairing
element_compatibility_score(water, fire, 22) :- !.
```

Excerpt from `proLog/synastry.pl` – a genuinely weighted, multi-factor scoring rule with cut-guarded lookup clauses. Thresholds and component scores shown are the rebalanced values (see §4.5).

5.5 Entry Point and Module Loading

```
:- use_module(zodiac).
:- use_module(tarot).
:- use_module(zodiac_profile).
:- use_module(card_selection).
:- use_module(reading_analysis).
:- use_module(compatibility).
:- use_module(synastry).
:- use_module(horoscope_rules).
:- use_module(spread_rules).
:- use_module(recommendation_rules).
:- use_module(reasoning).
:- use_module(history_analysis).

% Get zodiac info
zodiac_info(Sign, Info) :-
    zodiac(Sign),
    element(Sign, Element),
    modality(Sign, Modality),
    ruling_planet(Sign, Planet),
    zodiac_traits(Sign, Traits),
    zodiac_date_range(Sign, DateRange),
    zodiac_symbol(Sign, Symbol),
    Info = zodiac_info{
        sign: Sign, element: Element, modality: Modality,
        ruling_planet: Planet, traits: Traits,
        date_range: DateRange, symbol: Symbol
    }.
```

Excerpt from `prolog/main.pl` – single load point for all 12 modules plus convenience queries used directly by `PrologService`.

5.6 Reasoning Trace Format

Every module returns machine-readable trace entries the frontend renders as a step-by-step explanation of the inference:

```
[
  {
    "rule": "element(aries, fire)",
    "input": "aries",
    "result": "Aries belongs to fire element",
    "explanation": "Fire element defines personality style"
  }
]
```

6. Current Limitations

- **Precise chart requires birth time and place.** Sun, Moon, Ascendant, and Midheaven are computed exactly via the Swiss Ephemeris when birth time and location are supplied; without a location the system still falls back to a Sun-sign-only approximation, since Moon/Rising cannot be derived from date alone.
- **Western tropical zodiac boundaries only.** Sidereal/Vedic zodiac systems and alternative astrological traditions are out of scope.
- **Static trait vocabulary.** Strength/challenge phrasing is drawn from a fixed `trait_strength/2` / `trait_challenge/2` table; it does not adapt to combinations of traits beyond simple aggregation.
- **LLM dependency for narrative depth.** Rich, varied prose depends on OpenRouter availability; the deterministic fallback is functionally correct but noticeably more templated and repetitive.
- **Localization limited to two languages.** The UI, static content (signs, cards, spreads, horoscope themes), and LLM-generated narration (readings, chat, birth chart explanations) are available in English and Myanmar; the underlying Prolog knowledge base itself – atoms, predicate names, rule identifiers – remains English-only internally, and no additional languages are supported yet.
- **No user accounts / multi-tenant isolation.** Reading history is stored in a local SQLite database without authenticated, per-user separation.
- **Compatibility model is symbolic, not empirical.** The weighted scoring formula encodes traditional astrological symbolism; it is not validated against psychological or relationship-outcome data, and is presented purely as a reflective tool.
- **Question classification is heuristic.** Category detection uses keyword/pattern-based NLP rather than a trained classifier, so ambiguous or multi-topic questions can be mis-categorized.

7. System Implementation (UI Design)

7.1 Application Pages

Route	Purpose
/dashboard	Landing overview – quick access to readings, horoscope, and recent history.
/zodiac	Browse all 12 signs; view element, modality, ruling planet, traits, strengths and challenges.
/birth-chart	Enter birth date/time and, optionally, birth place for a precise Swiss Ephemeris-based natal chart – Sun, Moon, Ascendant, Midheaven and planetary degrees rendered on an interactive chart wheel, with an AI-narrated bilingual explanation. Falls back to a date-only Sun-sign profile without a birth place.
/tarot	Browse the 78-card deck; view keywords and upright/reversed meanings.
/reading	Ask a question, then either let the app draw a recommended (or custom 1–10-card) spread at random, or hand-pick specific cards straight from the deck; view the AI-narrated interpretation either way.
/horoscope	Generate a themed daily horoscope from sign and mood.
/compatibility	Compare two zodiac signs and view the weighted score breakdown and explanation.
/chat	Conversational AI assistant grounded in Prolog facts and reading context.
/reasoning	Dedicated reasoning-trace explorer – visualizes the rule chain behind a result.
/history	Review, revisit, and delete past readings and chat sessions.
/analytics	Aggregate statistics across a user's reading history.
/scan	Photograph or upload a physical tarot card; a vision-capable LLM identifies it (with a local keyword fallback), then returns its meaning and an AI-generated explanation.
/about	Project description and ethical framing.

A persistent `ProfileSettingsModal`, reachable from any page, lets the user save a name, birth date/time/place, and preferred zodiac sign for reuse across the reading, horoscope, and birth-chart forms; a language switcher (English / Myanmar) in the top bar is likewise available on every page and its selection persists across sessions.

7.2 Key UI Components

- `TarotCard` – renders card art, name, orientation, and meaning.
- `CardPicker` – searchable/filterable deck browser for manual card selection, with a reorderable selection tray (up to 10 cards) supporting per-card orientation toggling.
- `ZodiacBadge` / `ElementBadge` – compact visual tags for sign and element.
- `ReasoningStep` – renders one entry of a Prolog reasoning trace (rule, input, result, explanation).
- `NatalChartWheel` – SVG chart wheel plotting Sun/Moon/planet positions and sign divisions from a calculated birth chart.
- `BirthChartExplanation` – renders the AI-narrated, bilingual natal chart breakdown returned by `/api/birth-chart/explain`.
- `ProfileSettingsModal` – edits and persists the user's saved profile (name, birth details, preferred sign) used to pre-fill other pages.
- `LanguageSwitcher` – toggles the UI locale between English and Myanmar; shown in the top bar on every page.
- `MarkdownText` – renders lightweight Markdown (bold, italics, lists) from LLM-generated responses safely as HTML.
- `LoadingSpinner`, `StarField`, `Navigation` (top bar), `Sidebar` (primary link navigation) – shared visual/interaction chrome.

State is managed through dedicated Svelte stores – `Profile`, `Reading`, `Chat`, and `Loading` – with a typed API client that centralizes error handling for all backend calls, and an `i18n` locale store that drives every translated string in the app.

7.3 REST API Surface

```
GET /api/health
GET /api/zodiac
GET /api/zodiac/{sign}
GET /api/zodiac/{sign}/profile
GET /api/tarot
GET /api/tarot/{card}
POST /api/tarot/draw { count, spread_type }
POST /api/reading/analyze { question, zodiac_sign, spread_type, card_count }
POST /api/reading/analyze-selected { question, zodiac_sign, cards: [{name,
is_reversed}] }
POST /api/reading/generate
POST /api/horoscope/generate { zodiac_sign, mood }
POST /api/compatibility/analyze { sign1, sign2 }
POST /api/compatibility/synastry { sign1, sign2 }
POST /api/chat { message, zodiac_sign, current_reading }
GET /api/history
GET /api/history/{id}
DELETE /api/history/{id}
GET /api/analytics
POST /api/birth-chart/calculate { year?, month, day, hour, minute, latitude?,
longitude? }
POST /api/birth-chart/explain { chart, locale }
POST /api/tarot-scan/identify (multipart image upload, max 10MB)
POST /api/tarot-scan/explain { card_id, is_reversed? }
GET /api/topics
GET /api/topics/{topic}
POST /api/topics/classify { question }
```

Base URL: `http://localhost:8000/api` (from `docs/api.md`). `/api/birth-chart/calculate` returns a date-only Sun-sign profile when `year` / `latitude` / `longitude` are omitted, and a precise, degree-accurate chart (via the Swiss Ephemeris) when they are supplied.

7.4 Backend Service Layer

FastAPI is organized into `api` (route handlers), `services` (`PrologService`, `TarotService`, `TarotAPIClient`, `HoroscopeService`, `AnalyticsService`, `ChatService`, plus the function-based `openrouter_service`, `ephemeris_service`, and `birth_chart_explanation_service` modules), `schemas` (Pydantic request/response models), `models`, and `database` (SQLite access). `PrologService` is the single bridge between Python and SWI-Prolog, exposing typed methods for every predicate in each of the four reasoning modules (e.g. `get_zodiac_profile`, `select_eligible_cards`, `analyze_reading`, `analyze_synastry`). `ephemeris_service.compute_natal_positions` wraps `pyswisseph` to compute exact ecliptic positions (run off the event loop via `asyncio.to_thread` to keep the API responsive under concurrent requests), and

`birth_chart_explanation_service.generate_birth_chart_explanation` turns a calculated chart into a bilingual, LLM-narrated explanation, while `openrouter_service` additionally exposes a vision-capable call path (`scan_tarot_image`) used by `/api/tarot-scan/identify` to recognize a photographed card.

7.5 Visual/UX Design Notes

- Dark, star-field themed interface (via the `StarField` component) consistent with the astrology/tarot subject matter.
- Reasoning is never hidden: every generated result links to a reasoning-trace view so users can see *why* the system produced that answer.
- Responsive layout built with Tailwind CSS utility classes; component styles kept local to each Svelte component.
- Card art is served through an image-resizing proxy (`wsrv.nl`) rather than linked directly to the source tarot API: the source serves full-resolution scans ($\sim 1150 \times 1900$ px, ~ 1.1 MB each) for every card, which on a slow, congested, or metered network can stall or fail before a thumbnail-sized image ever needs that much data – the proxy re-encodes each card as a right-sized WebP (e.g. a 300px-wide thumbnail drops from ~ 1.1 MB to ~ 45 KB) and edge-caches it, which also incidentally routes around networks that block the source host directly.

8. Conclusion

AstroLogic demonstrates that a large, traditionally free-text domain – astrology and tarot – can be modeled as a structured, rule-based reasoning system. By encoding zodiac and tarot knowledge as declarative Prolog facts and layering four purpose-built reasoning modules on top (profile derivation, card selection, reading analysis, and synastry scoring), the project produces results that are **explainable and reproducible**: every output can be traced back to the specific facts and rules that generated it. The LLM is deliberately confined to natural-language presentation, which keeps the system's actual "intelligence" auditable rather than opaque, and provides a robust deterministic fallback path. Combined with a modern SvelteKit interface and a documented FastAPI backend, AstroLogic delivers a coherent, end-to-end example of symbolic AI applied to an engaging, user-facing product – and shows in practice how classical logic-programming techniques (fact bases, Horn-clause rules, weighted scoring, cut-guarded lookup) remain a powerful, transparent complement to modern generative AI.

9. Further Extension

- **Planetary house placements.** Extend the natal chart beyond sign/degree positions to compute house cusps and placements (already have Ascendant/Midheaven as a foundation), enabling house-based interpretation.
- **Trained question classifier.** Replace the heuristic category detector with a lightweight trained NLP model for more accurate spread recommendation.
- **Personalization over history.** Use `history_analysis.pl` more extensively to surface long-term patterns (recurring themes, sign trends) back to the user through the Analytics page.
- **Additional languages.** Extend the existing English/Myanmar i18n layer with further languages, now that the translation dictionary, locale-aware LLM prompts, and static content structures are already in place.
- **User accounts and multi-tenancy.** Add authentication so reading history is securely scoped per user.
- **Alternative zodiac systems.** Add optional sidereal/Vedic rule sets alongside the existing Western tropical rules.
- **Expanded compatibility model.** Incorporate additional symbolic factors (e.g. house overlays, aspect angles) once full chart data is available.
- **Offline/edge deployment.** Package the Prolog engine and a small local LLM together for an offline-capable version of the app.

10. References

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